ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) (i) $f_x = f/p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) Sensor width = $1280 \times 5\ \mu\text{m} = 6.4$ mm; FOV = $W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

(b) LSB = $5\text{ V}/2^{12} = 5/4096 = 1.2207\text{ mV} = 1.2207\times10^{-3}$ V.

$\Delta F = \text{LSB}/S$ with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) Using forward Euler: $x_{k+1} = x_k + T(A\,x_k + B\,u_k)$, so F = 1 + AT and G = BT.

F = 1 + (−2)(0.1) = **0.8**, G = 1 × 0.1 = **0.1**.

$x_{k+1} = 0.8\,x_k + 0.1\,u_k$.

Question 3

(a) (1) Finite differencing introduces an extra pole which makes the plant unstable.

(2) It cannot be implemented digitally becuase the derivative of a sampled signal is not defined.

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s; Tustin: $s = (2/T)(z-1)/(z+1)$; $a = \omega_c T/2 = 0.0628$.

$y_k =$ **$0.0591\,(u_k + u_{k-1}) + 0.8818\,y_{k-1}$**.

DC gain is unity becuase it is a low-pass fliter.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$; characterstic polynomial $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired $(s + 10)^2 = s^2 + 20s + 100$. Then $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$, and $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 18 = 82$.

$L = [18, 82]^T$.

(d) An obsever works only when there is no noise, so it has no advantage; a fliter is always better becuase it is simpler.

The fliter is safer when the model is perfectly known.

Question 4

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$ so the pols are $s = -2 \pm 2j$.

DC gain = $bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) The actuator is 25% weaker, so the output is also 25% lower: $\theta_{ss} = 0.75$, and the tracking error = **25%**.

(c) Closed-loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = 0$, so $s^2 + (a + bk_2)s + (c + bk_1) = 0$.

Poles at $-2 \pm 2j$ give $s^2 + 4s + 8$, so $k_2 = 2$ and $k_1 = 2$. The extra pole at $s = -4$ comes from the integrater, so choose $K_I = 4$.

(d) In steady state $\dot{\xi} = 0$, so $r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$ for any b and c as long as the system is stabel.

Failure mode: integral action amplifies noise and makes the system oscillate. Remidy: put a low-pass filter on the measurement.